



Enhancing programming learning experience using quarto-live

Applied in teaching Clinical Data Validation using R

Jeremy Selva | Saturday 26 September 2026

Restricted, Non-Sensitive

PATIENTS. AT THE HE^{RT} OF ALL WE DO.®



Singapore General Hospital



Changi General Hospital



Sengkang General Hospital



Eastern General Hospital



KK Women's and Children's Hospital



National Cancer Centre Singapore



National Dental Centre Singapore



National Heart Centre Singapore



National Neuroscience Institute



Singapore National Eye Centre



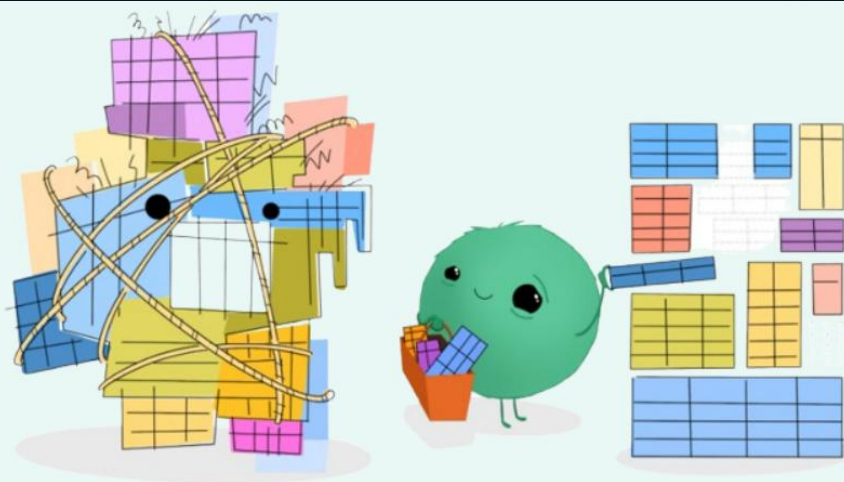
SingHealth Community Hospitals



Polyclinics SingHealth

Who am I

Research Officer from National Heart Centre Singapore who collects, cleans and harmonises clinical data.



Taming the Data Beast from "Cleaning Medical Data with R" workshop by Shannon Pileggi, Crystal Lewis and Peter Higgins presented at R/Medicine 2023. Illustrated by Allison Horst.

Research Work

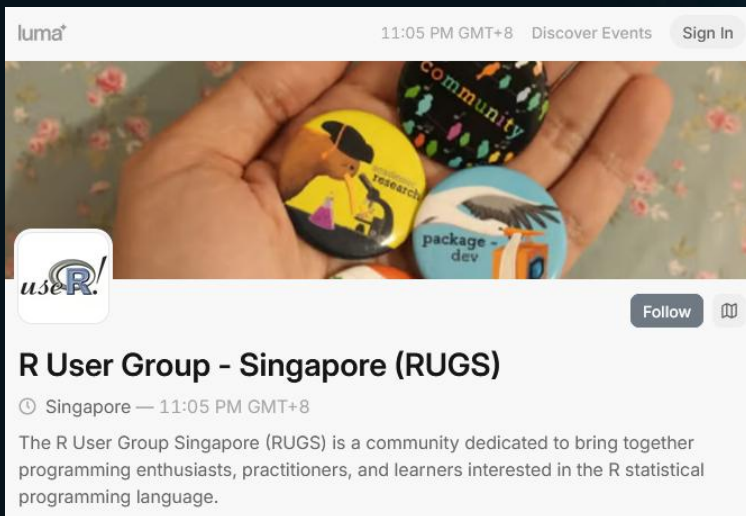
Global Pretest Probability Study of Coronary Artery Disease

<https://clinicaltrials.gov/study/NCT05722145>

- Consolidate a global registry of patients referred for CT scan to compare existing pretest probability models to predict CAD and major adverse cardiovascular outcomes.
- Currently, ingested **~80,000** patients across **19 countries**.
- Harmonisation of clinical data, race/ethnicity, coronary calcium scores, coronary CT stenosis results, outcomes



R User Group Singapore



<https://luma.com/rugs>

R related talks

Tackling Formatted Tabular Data from Excel

10th July 2024

Jeremy Selva 
@JauntyJJS 
<https://jeremy-selva.netlify.app> 
For useR! 2024 

<https://jauntyjjs.github.io/useR-2024/>

Retrospective clinical data harmonisation Reporting Using R and Quarto

Jeremy Selva 
@JauntyJJS    
<https://jeremy-selva.netlify.app> 
For Harvest Analytics Together (HAT) 2025 
14th November 2025



https://jauntyjjs.github.io/hat_2025

R Package Contributions

RiskScorescvd 0.3.1 Reference Articles Changelog

Search for

RiskScorescvd



Chest pain? Calculate your cardiovascular risk score.

The goal of RiskScorescvd R package is to calculate the most commonly used cardiovascular risk scores. Original research publication can be found here: <https://openheart.bmj.com/content/11/2/e002755>

We have developed nine of the most commonly used risk scores with a dependency (ASCVD [PooledCohort]) making the following available:

- HEART
- EDACS
- GRACE 2.0
- TIMI
- SCORE2/OP
- SCORE2-Diabetes
- SCORE2-CKD
- SCORE2-Asia
- PCE (ASCVD)

NOTE: Troponin I values should be used. Additional functions for Troponin T are under development

Links

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)

License

- [Full license](#)
- [MIT](#) + file [LICENSE](#)

Citation

- [Citing RiskScorescvd](#)

Developers

Daniel Perez-Vicencio
Author, maintainer

Dimitrios Doudesis
Author

Alexander JF Thurston
Author

Jeremy Selva
Author

pretestcad 1.2.0 Reference Changelog

Search for

pretestcad



R package used to calculate different **PreTest** Probability (PTP) scores for obstructive **Coronary Artery Disease** (CAD).

As diagnosis of CAD involves a costly and invasive coronary angiography procedure for patients, having a reliable PTP for CAD helps doctors to make better decisions during patient management. This ensures high risk patients can be diagnosed and treated early for CAD while avoiding unnecessary testing for low-risk patients.

Table of Content

- [Installation](#)
- [Currently available pretest probability scores](#)
- [Getting Started](#)

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- [Report a bug](#)

License

- [Full license](#)
- [MIT](#) + file [LICENSE](#)

Citation

- [Citing pretestcad](#)

Developers

Jeremy Selva
Author, maintainer

Risk of CVD : <https://dvicencio.github.io/RiskScorescvd/>

Risk of CAD : <https://jauntyjjs.github.io/pretestcad/>

Why clinical data validation **matters**

- GenAI can produce codes to do data analysis and generate synthetic data **faster and cheaper** than manual work.
- Its term of use reminds users that **outputs** may **contain mistakes** and **advise "humans in the loop"** in AI-related workflows



Hand drawn ai alignment illustration by **magnific**

Data validation is hard



ADVERSE EVENT (AE) DATA – CAN YOU SPOT THE DATA ISSUES?



Review the AE dataset below and identify any potential data quality issues.

STUDYID	USUBJID	AESEQ	AETERM	AEDECOD	AEBODSYS	AESTDAT	AEENDAT	AESEV	AESEV	AEREL	AEOUT	AEACN	AECONTRT	AESDTH	AESHOSP	AESCONG	AESLIFE	AESDISAB	AESMIE
AS0017	AS0017-001-0001	1				01-08-2026	09-08-2026	No	Mild	No	Recovered/ Resolved	Dose not changed	No	No	No	No	No	No	No
AS0017	AS0017-001-0001	2	Nausea	Nausea	Gastrointestinal disorders	07-08-2026		No	Mild	No	Recovered/ Resolved	Dose not changed	Yes	No	No	No	No	No	No
AS0017	AS0017-001-0001	3	Vomiting	Vomiting	Gastrointestinal disorders	08-2026	24-07-2026	Yes	Moderate	No	Not recovered/ Not resolved	Drug interrupted	Yes	No	No	No	No	No	No
AS0017	AS0017-001-0002	1	High blood pressure	Hypertension	Vascular disorders	10-07-2026		No	Severe	Yes	Not recovered/ Not resolved	Drug permanently discontinued	Yes	Yes	No	No	No	No	No
AS0017	AS0017-001-0002	2	Skin rash	Rash	Skin and subcutaneous tissue disorders	21-07-2026	31-07-2026	No	Mild	No	Recovered/ Resolved	Dose not changed	No	No	No	No	No	No	No
AS0017	AS0017-001-0003	1	Dizziness	Dizziness	Nervous system disorders	15-08-2026	14-08-2026	No	Mild	Yes	Recovering/ Resolving	Dose not changed	No	No	No	No	No	No	No
AS0017	AS0017-001-0003	2	Cough			2026	27-08-2026	Yes	Moderate	No	Not recovered/ Not resolved	Dose not changed	Yes	No	Yes	No	No	No	No



Amit Satra • 2nd
SME – Statistical ...
1d •

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👍 39
💬 9
🔄
📌
🌐



😊 GIF 🖼️

Most relevant ▾



Chen Wei Oh • 1st
Statistical analyst for clinical trial. I help pharma ...
1d •

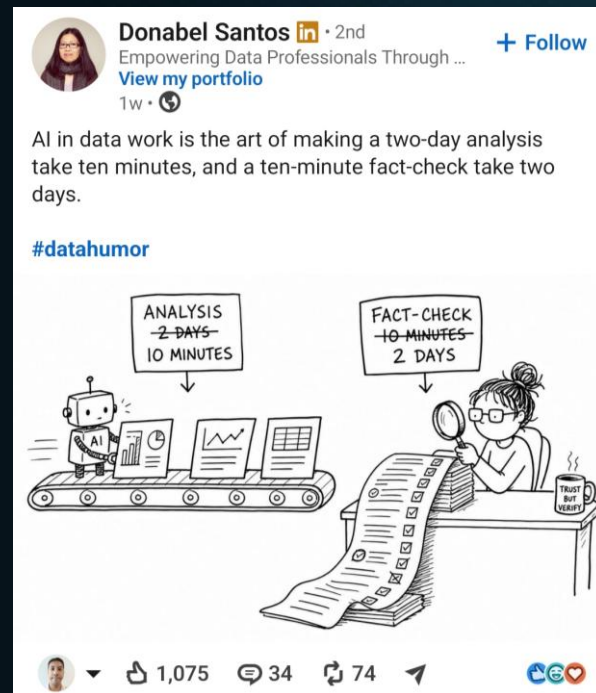
Aeendt before aestdt. AESER = Y but all SAE criteria = N

Linkedin post by **Amit Satra**.

Data validation is **hard**

Data verification process is **time consuming** and **tedious**.

- **Harder to spot** than text errors by humans and machine
- Lead to **larger** clinical or operational consequences.

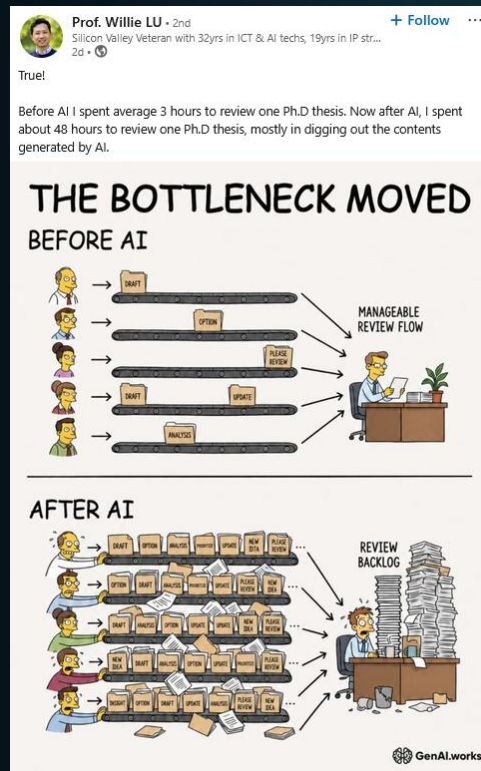


Linkedin post by **Donabel Santos**.

The human-in-the-loop **bottleneck**

GenAI outputs **scale faster** than human verification can keep up

- If someone reviews 10 AI outputs/day, they can genuinely check.
- If the someone “oversees” 10,000 outputs/day, meaningful review becomes impossible and turns into **rubber-stamping**.



Linkedin post by Prof. Willie LU.

The human-in-the-loop **bottleneck**

A common misunderstanding is treating “human-in-the-loop” as **merely having a human present**, without systematic cross-checking and validation.

The screenshot shows a Zoom webinar interface. The top bar includes browser tabs for 'Hands Dirty First', 'Lobby | Zoom Webinars & Events', and another 'Hands Dirty First' tab. The address bar shows the URL 'pzhang724.github.io/GenAI-in-Pharma-2025/#/but-am-i-really-in-the-loop'. The slide content is as follows:

But — am I really in the loop?

- The point is not how fast I generate those.
- One uncomfortable truth: after generating the data, I **know almost nothing** about it.
- I can't be sure the **code** is right or the **data** is right.
- I only know: the **spec is written**, **code generated**, **no errors**, **data produced** — and I'm a human in the loop.
- Is that what “human-in-the-loop” should mean? What are we missing?
- The missing piece: an **immediate human review / approve / authorize** step to ensure correctness.

At the bottom of the slide, it says 'GenAI in Pharma 2026 — Virtual' and '16 / 25'. Below the slide, the text 'Practical Notes from Before, During, and After Building Agentic Workflows - Mia Chen' is displayed.

Practical Notes Before, During, and After Building Agentic Workflows by Mia Chen, presented at **GenAI in Pharma 2026**.

Validation strategies

There is a pressing need to be **aware of ways** to make the data verification workflow more systematic and efficient.

- One approach is to let another GenAI model do the verification and reporting process (Require **skills.md**)
- Identify/Understand/Create **deterministic** methods to efficiently **validate** data based on domain knowledge and **report** issues clearly



Action process directions performance verification icon by rawpixel.com on [Magnific](https://magnific.com).

Abundance Resources



[pointblank](#)
[checkmate](#)
[data.validator](#)



[Pandera](#)
[Patito](#)
[Dataframely](#)



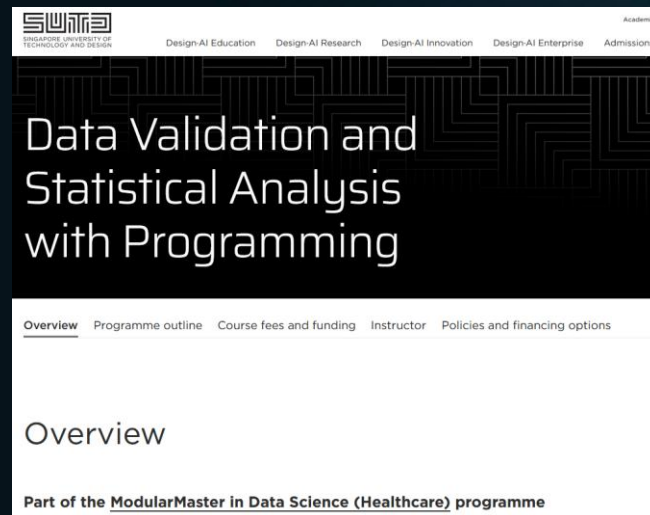
[SimpleValidations.jl](#)

Abundance Resources

There's also growing number of workshops from tertiary institutions.



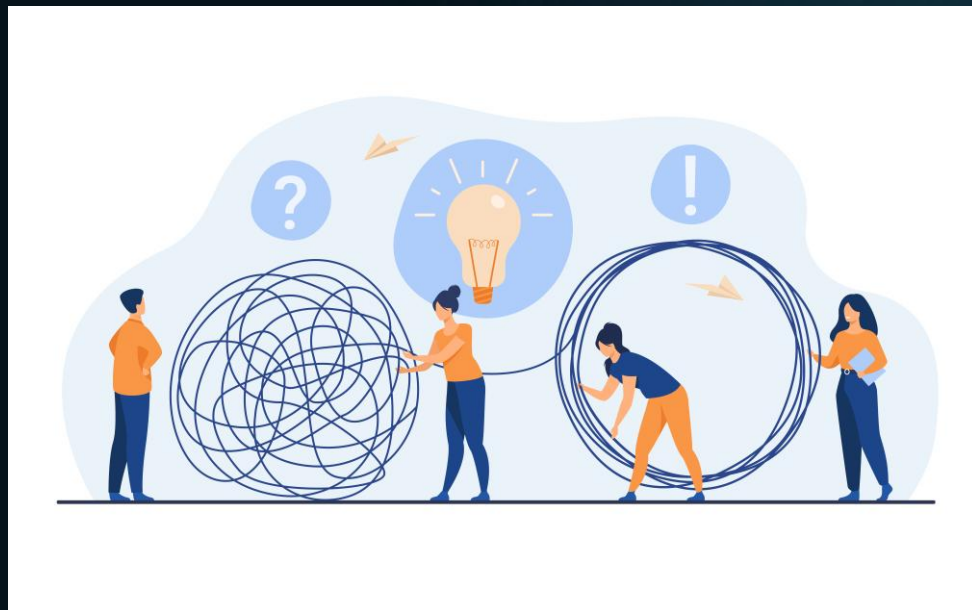
<https://www.duke-nus.edu.sg/cbds/events-news/events/detail/index/measuring-what-matters-practical-guide-to-selecting---validating-patient-reported-outcome-measures>



<https://www.sutd.edu.sg/academy-course/healthcare-data-validation-and-statistical-analysis-with-programming>

Adoption gap

- Workshops are **resource-intensive**, hard to scale to a large audience quickly.
- Workshops **cannot cover** specific **libraries** for **niche**/complicated applications.



Team of crisis managers solving businessman problems by pch.vector on Magnific.

Adoption gap

Clinicians encounter **low** knowledge retention and find it **hard to reapply** knowledge needed more than six months later.

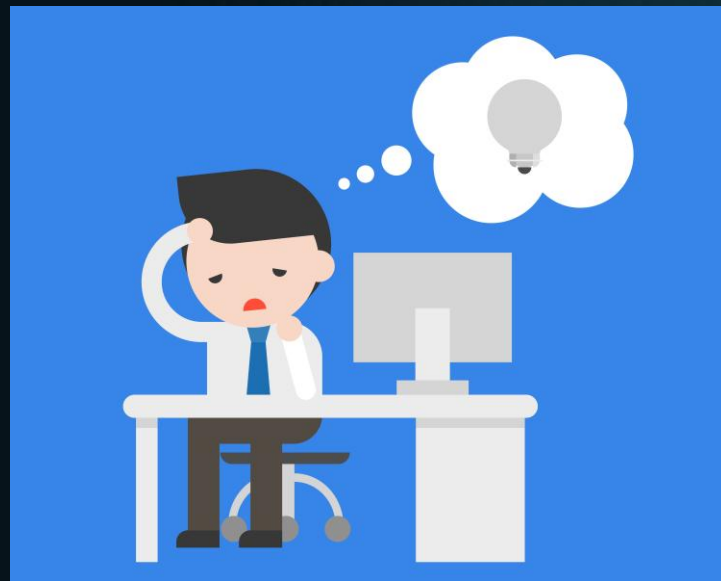
- “Open Rstudio and then Open R script... and then ..? and then ???”
- “I forget liao, notes all give back to teacher.”



Wind-up businessman by Amonrat Rungreangfangsai.

Adoption gap

- Clinicians struggle to **navigate complex documentation** of programming libraries that they never see before.
- Can be **demoralising** to invest time only to find a **tool isn't suitable**.



Businessman sitting at office and feeling dull because lack of idea by **Amonrat Rungreangfangsai**.

Advantages of Online interactive coding tutorials

Remove installation barriers so users can focus on learning rather than setup in different computers.

Users can try different inputs to see how functions behaves

Learning pace is flexible for users to return anytime for refreshers, improving knowledge retention.

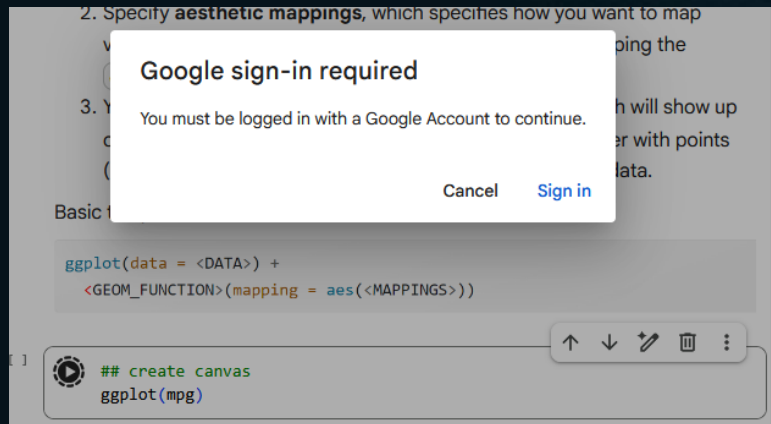
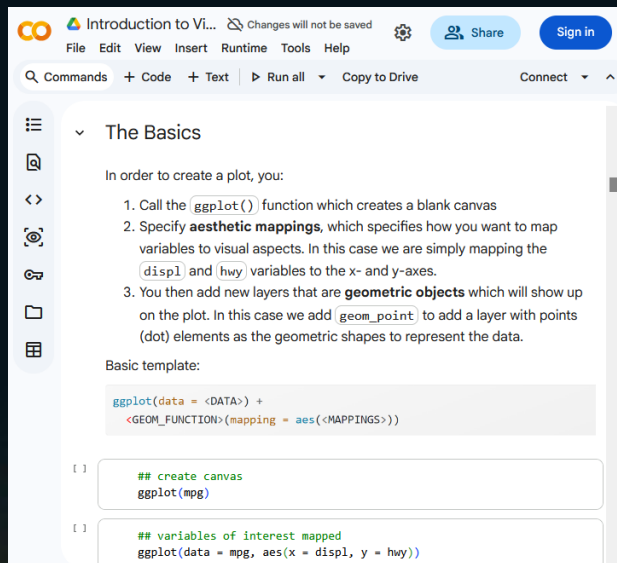
```
1 # Try a Dataquest exercise!  
2 # We loaded in the Fortune 500 list  
3 # Hit Run Code to show average revenue  
4  
5 avg_revenue = mean(revenues)  
6 print(f"Average revenue: ${avg_revenue}M")
```

Run Code

Interactive code block from Dataquest.
<https://www.dataquest.io/>

Jupyter Notebook

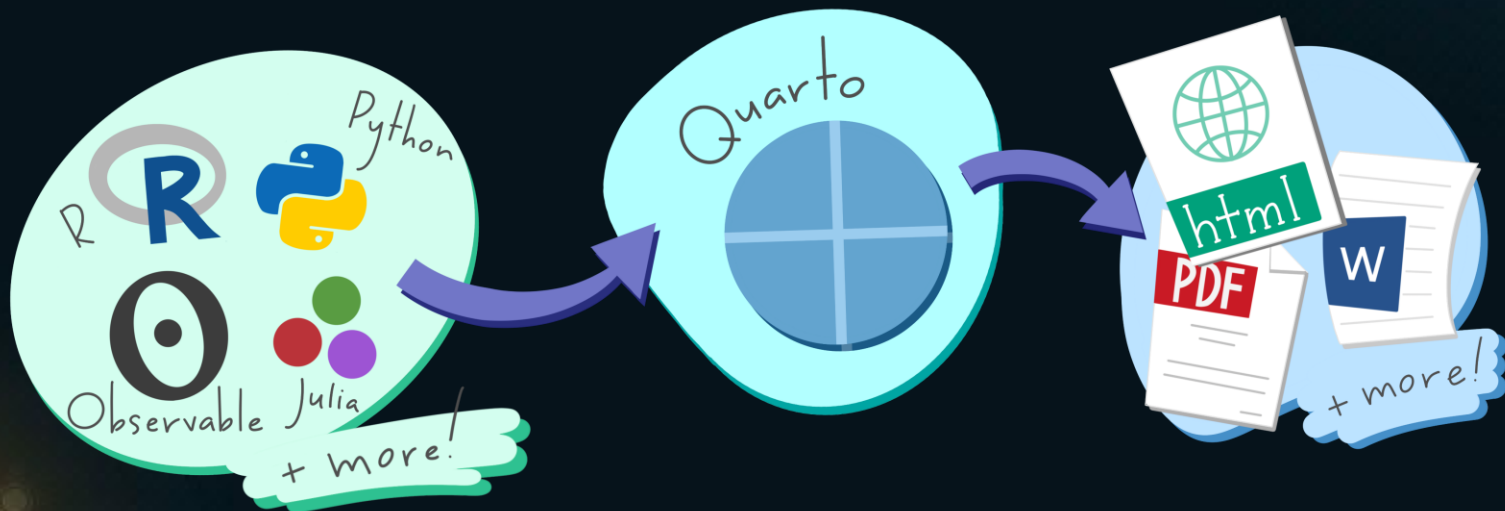
- Need a server environment (Google Colab) to run it.
- Requires Google Account (Gmail) to run the code.



Introduction to Visualization with ggplot2 in Google Colab

Quarto

Quarto is a scientific and technical publishing system that combines markdown text with executable code (R/Python/Julia/JavaScript) to create professional reports, **websites**, and presentations.



Artwork from “Hello, Quarto” keynote by Julia Lowndes and Mine Çetinkaya-Rundel, presented at RStudio Conference 2022. Illustrated by [Allison Horst](#).

quarto-live



quarto-live is a Quarto extension that uses **WebAssembly** to run interactive R and Python code cells directly inside the reader's web browser, **without** requiring an active backend server.

Source

R Python

```
```{webr}
for (x in 1:5) {
 print(10 + x)
}
```
```

Output

R Python

R Code Start Over Run Code

```
1 ✓ for (x in 1:5) {
2   print(10 + x)
3 }
```

Source

R Python

```
```{pyodide}
for x in range(1, 6):
 print(10 + x)
```
```

Output

R Python

Python Code Start Over Run Code

```
1 ✓ for x in range(1, 6):
2   print(10 + x)
```

WebAssembly logo from: <https://webassembly.org/>

quarto-live preview from: https://r-wasm.github.io/quarto-live/getting_started/editor.html

Examples

Use R and Python to do plotting

- https://r-wasm.github.io/quarto-live/getting_started/plotting.html

Plotting and Graphics

Dynamic data visualisation for interactive code blocks

The `quarto-live` extension prepares the R and Python environment with WebAssembly-compatible graphics devices, and so plotting should work out of the box.

R

Python

R Code

Start Over

Run Code

```
1 library(ggplot2)
2 ggplot(mtcars, aes(x = wt, y = mpg, group = am)) +
3   geom_point(aes(color = cyl, shape = cyl)) +
4   geom_smooth(method = lm, formula = y ~ x) +
5   ggtitle("Motor Trend Car Road Tests 🚗") +
6   scale_shape_binned()
```


Examples

Explore Neural Networks Interactively with Quarto Live!

- https://rtichoke.netlify.app/posts/quarto_live.html

Building a Basic Neural Network

R Code

Start Over

Run Code

```
1 # Build a neural network with 3 nodes in the hidden layer
2 nn_model <- neuralnet(
3   formula = y ~ x1 + x2, # Predict y using x1 and x2
4   data = sample_data,    # The training dataset
5   hidden = c(3),         # One hidden layer with 3 neurons
6   linear.output = FALSE, # Use sigmoid activation (for classification)
7   threshold = 0.01       # Convergence threshold
8 )
9
10 # Visualize the network
11 plot(nn_model, rep = "best")
```

On this page

What is Quarto Live?
Creating a Simple Dataset
Understanding Neural Networks
Building a Basic Neural Network
Experimenting with Different Architectures
Evaluating Model Performance
Additional Resources



Examples

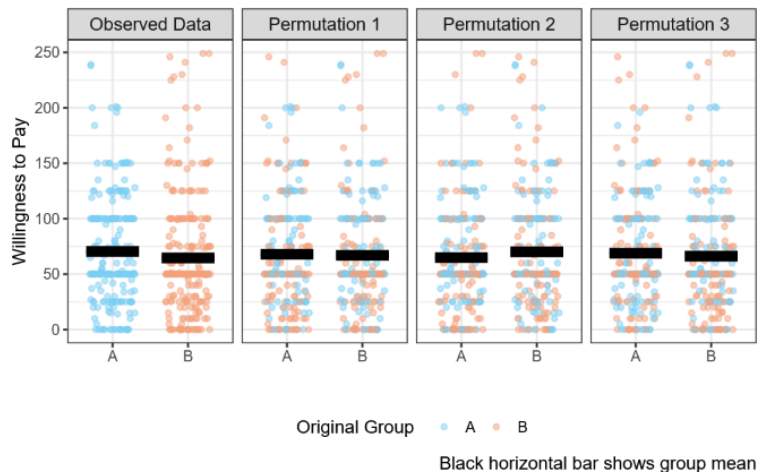
Null Hypothesis Significance Testing via Permutation Tests

- <https://kimberlyhenry.github.io/UseR2025/>



Visualizing Observed and Permuted Group Assignments

First panel shows the real data; remaining panels show shuffled labels



Take a peek at the raw data and first three shuffled datasets

The graphic below shows how permutation works. The first panel displays the original data with the true group assignments, and the next three panels illustrate the first three shuffles of those assignments. Notice that after shuffling, the group labels no longer carry real meaning — any patterns that emerge are purely due to chance.

R Code

↺ Start Over

▶ Run Code

```
1  
2 # add observed data as permutation 0  
3 observed_df <- my_groups |>
```

Demo



Clinical Data Validation in R

Validation Basics



Validation Advanced



R Programming



Clinical Data Validation in R

A guide on how to do clinical data validation using some R packages.

AUTHOR

Jeremy Selva

Preface

This book provides structured examples of ways to do clinical data validation using some R packages such as [pointblank](#).

Who is this for?

This is not a beginner's introduction to R or data programming in general, though you do not need to be an expert to understand its content. If you have some familiarity with tabular clinical data, most of the examples should make sense.

On this page

[Preface](#)

[Who is this for?](#)

[R Packages Used](#)

[R Platform Information](#)

[Credit](#)



https://jauntyjjs.github.io/clinical_validation_r/



HAT Workshop 20261021

Data Preparation



Table Making



HAT Workshop 20261021

Creating Clinical Summary Tables using gtsummary

AUTHOR

Jeremy Selva

Preface

This workshop aims to introduce [gtsummary](#) by creating clinical summary tables using the [Obstetrics and Periodontal Therapy Dataset](#) from [medicaldata](#). As the dataset is meant to be the source data for the journal [Treatment of Periodontal Disease and the Risk of Preterm Birth](#), we try to create the tables such that it matches the table content ([Table 1](#) and [Table 2](#)) from the paper as much as possible.

HAT 2026

Friday, 19 November 2026 (8.30am – 6pm) at Suntec Singapore Convention

- Registration fee: \$10
- Details: <https://www.cgh-hat.com.sg/>
- Sign up: <https://for.sg/hat2026-signup>



The banner features a friendly robot character with a screen for a face displaying a heart and the text "WELCOME! VITAL SIGNS GOOD!". The robot is holding a clipboard with a heart on it. The background shows industrial pipes and gears.

CGH H.A.T.

Harvest Analytics Together (H.A.T.) is an initiative by Changi General Hospital (CGH) to build a community of healthcare professionals and partners interested in leveraging data analytics to enhance patient care, improve work processes and support decision-making in healthcare.

Upcoming Events

| | | |
|--|--|---|
| <p>☆</p> <p>H.A.T. Incubator (Registration closed)</p> <p>Develop P.O.Cs for real-world healthcare problem statements.</p> <p>Find out more →</p> | <p>db</p> <p>Salesforce Data Viz Contest (submit by 7 Nov 26)</p> <p>Submit your best data visualisation for voting and stand a chance to win a prize!</p> <p>Find out more ↗</p> | <p>📅</p> <p>Workout Wednesdays</p> <p>Catch up on technical sharing sessions by Healthcare Analysts</p> <p>Find out more →</p> |
|--|--|---|

Let's Connect



<https://www.linkedin.com/in/jeremy-selva-085b9112a/>